



Erasmus KA1 Mobility Analysis

Students: **Opessio Daniele, Pasquino Riccardo, Valentino Raffaele**

Scientific Visualization

2025-2026

WORK FLOW



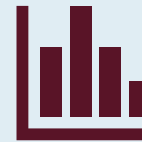
The Goal



The Dataset



**Data Cleaning and
Data Processing**



Data Visualization



**Conclusion and
Future Improvements**

The GOAL

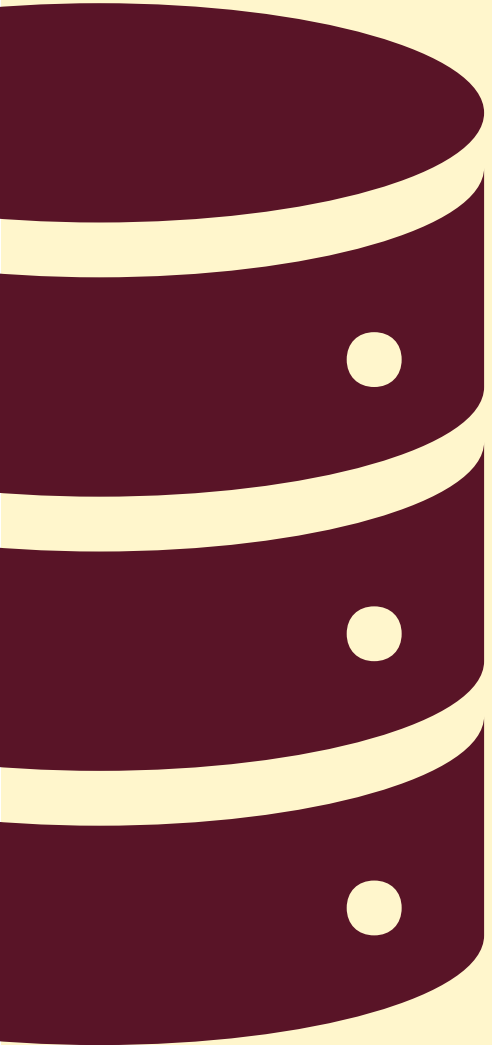


Erasmus+ is meant to foster cooperation, learning mobility and active participation among students throughout Europe.

How has it fared in the first decade of its existence?

Is the European Education Area coming to?

The DATASET



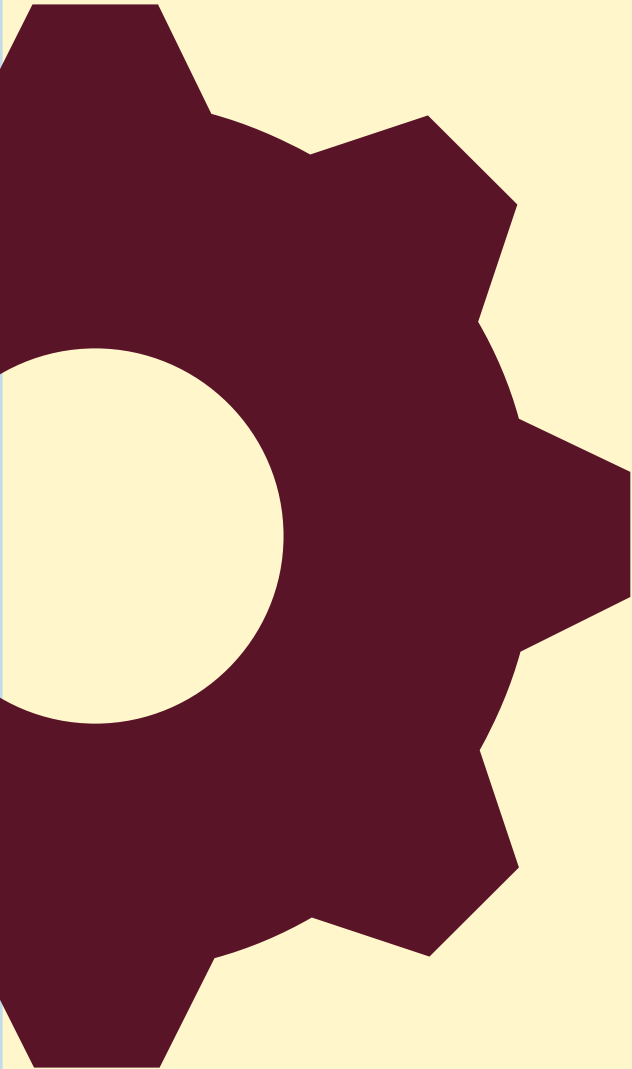
The relevant datasets can be found at the dedicated [European Commission page](#).

Each row contains information regarding different types of exchange, ranging from anonymized informations on the exchangee to informations on the receiving institution.

Each year had its own .xlsx file, thus they had to be merged into a single, consistent dataframe.

Naming inconsistencies had to be normalized

DATA CLEANING and DATA PROCESSING



- Filter on *Activity (mob)*: only university student mobility was kept, every other form of exchange was discarded.
- Removed useless columns and missing values.
- Map the *Field of Education* and *Organizations* labels to enable consistent analysis downstream.
- Made the Country naming scheme consistent across columns.

The resulting dataset shape is 3172958 x 14

Column Selection and Rationale

After combining and filtering, some columns of the resulting dataset were found to be either redundant or no longer needed. As such, they were safely dropped.

Column	Reason for removal
<i>Mobility Start Month, Mobility Start Year/Month</i>	Redundant with <i>Academic Year</i>
<i>Activity (mob)</i>	To filter non students out, all remaining rows are students.
<i>Field</i>	Redundant with <i>Field of Education</i>
<i>Participant Profile</i>	As all remaining rows are 'Learner' , it is now useless
<i>Actual Participants, Actual Participants (Contracted Projects)</i>	Not relevant for our analysis.
<i>Project Reference</i>	Not relevant for our analysis.

Column Cleaning and Standardization

Field of Education was unusable as-is due to inconsistent naming for the same universities.

To solve this issue, the content of the column was mapped to standardized labels via lookup table, leaving those not present in it unchanged.

0410 - Business and administration, not further defined /

041 - Business and administration →

Business and administration

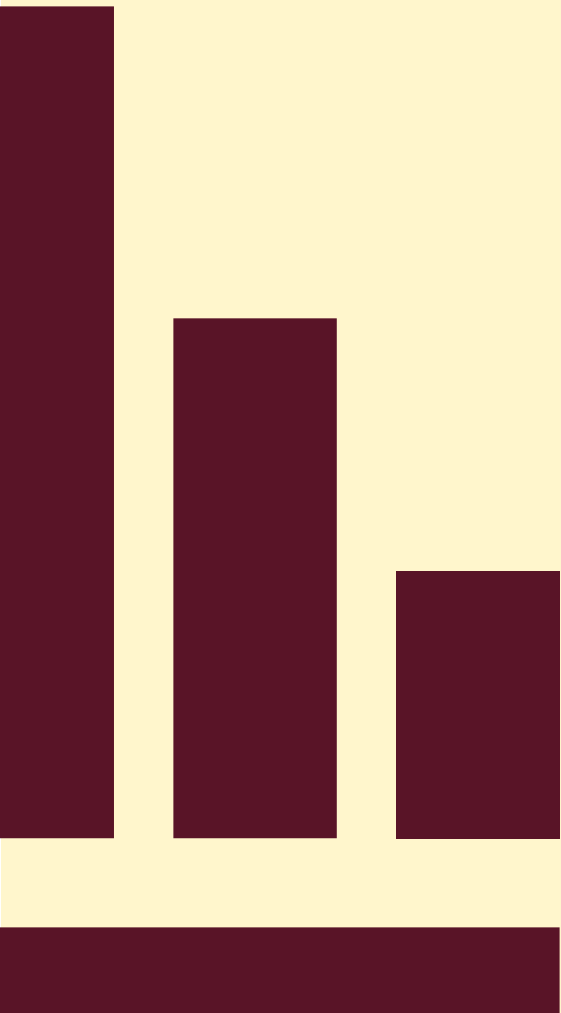
The same approach was used for the *Sending and Receiving Organizations* columns.

UMIL → Università degli Studi di Milano

Final Dataset

Academic Year	Mobility Duration	Field of Education	Participant Country	Education Level	Participant Gender	Fewer Opportunities	Participant Age	Sending Country	Sending City	Sending Organization	Receiving Country	Receiving City	Receiving Organization
2023	61	Business and administration	Austria	ISCED-6 - First cycle	Female	0	22	Austria	WIEN	WU	Germany	Munich	-
2023	101	Economics	Germany	ISCED-7 - Second cycle	Male	0	27	Austria	WIEN	WU	France	Toulouse	Université Toulouse Capitole
2023	121	Business and administration	Austria	ISCED-6 - First cycle	Female	0	22	Austria	WIEN	WU	France	Paris	-
2023	171	Business and administration	Germany	ISCED-6 - First cycle	Male	0	22	Austria	WIEN	WU	France	Paris	-
2023	86	Business and administration	Austria	ISCED-6 - First cycle	Male	0	23	Austria	WIEN	WU	Germany	Hamburg	-

DATA VISUALIZATION



Three levels of granularity:

1. Europe – European-wide trends.
2. Italy – Focus on Italian universities.
3. UniMi – How is our University performing.

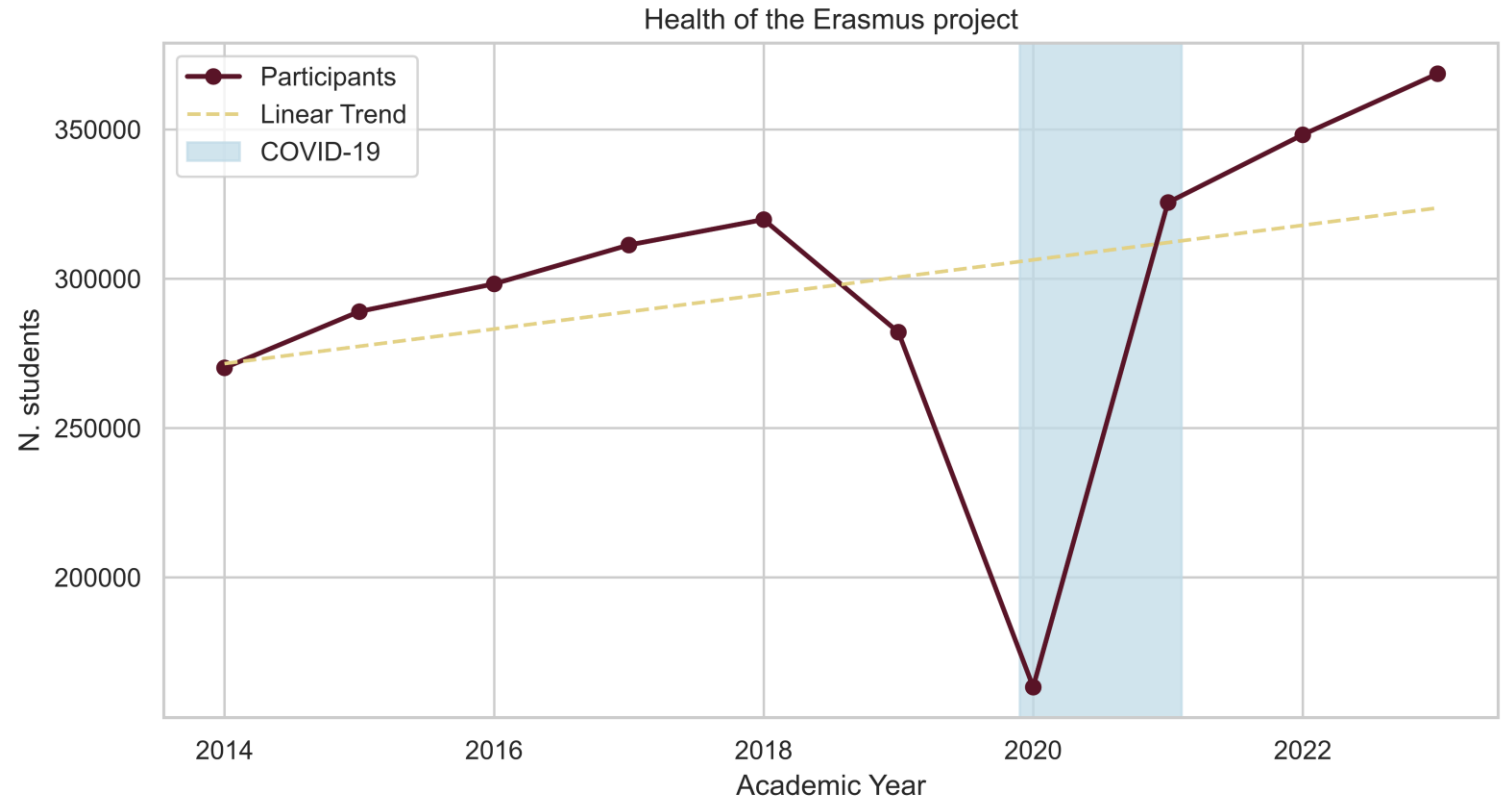
Ever closer to home @UniMi

EUROPE



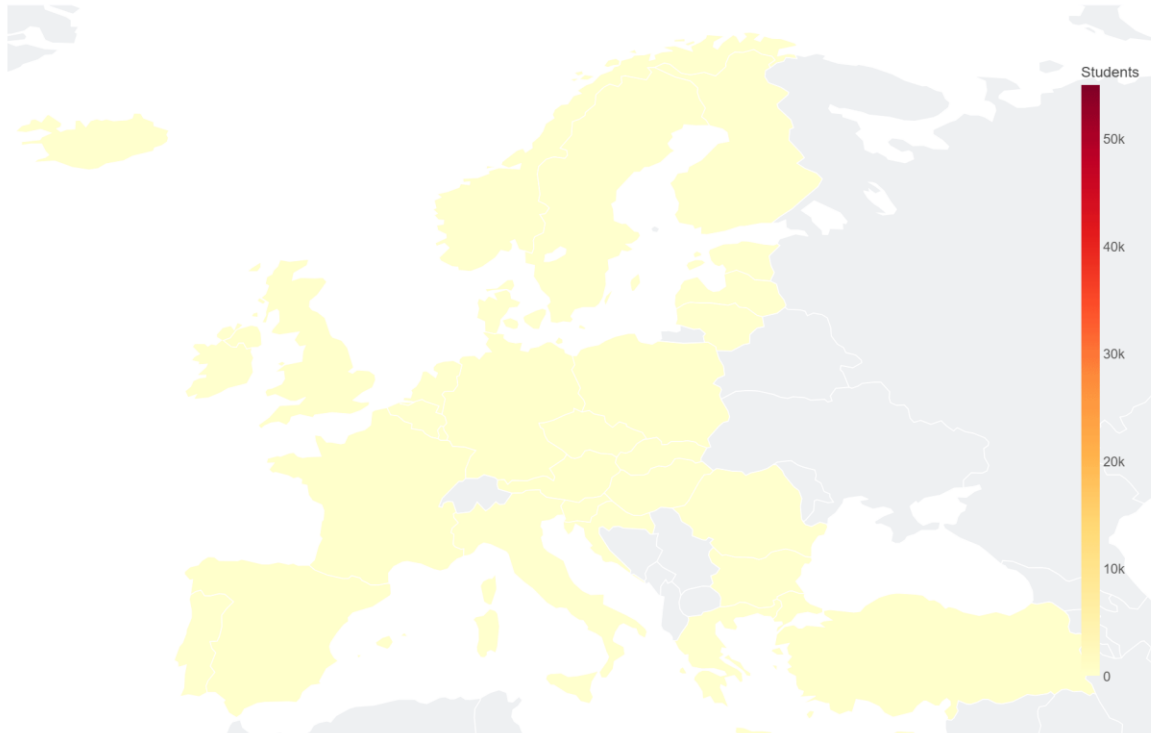
Health of Erasmus Project

Although Erasmus+ participation **fell sharply during the COVID-19 pandemic**, it bounced back quickly, showing an **even bigger interest in the programme afterwards**.

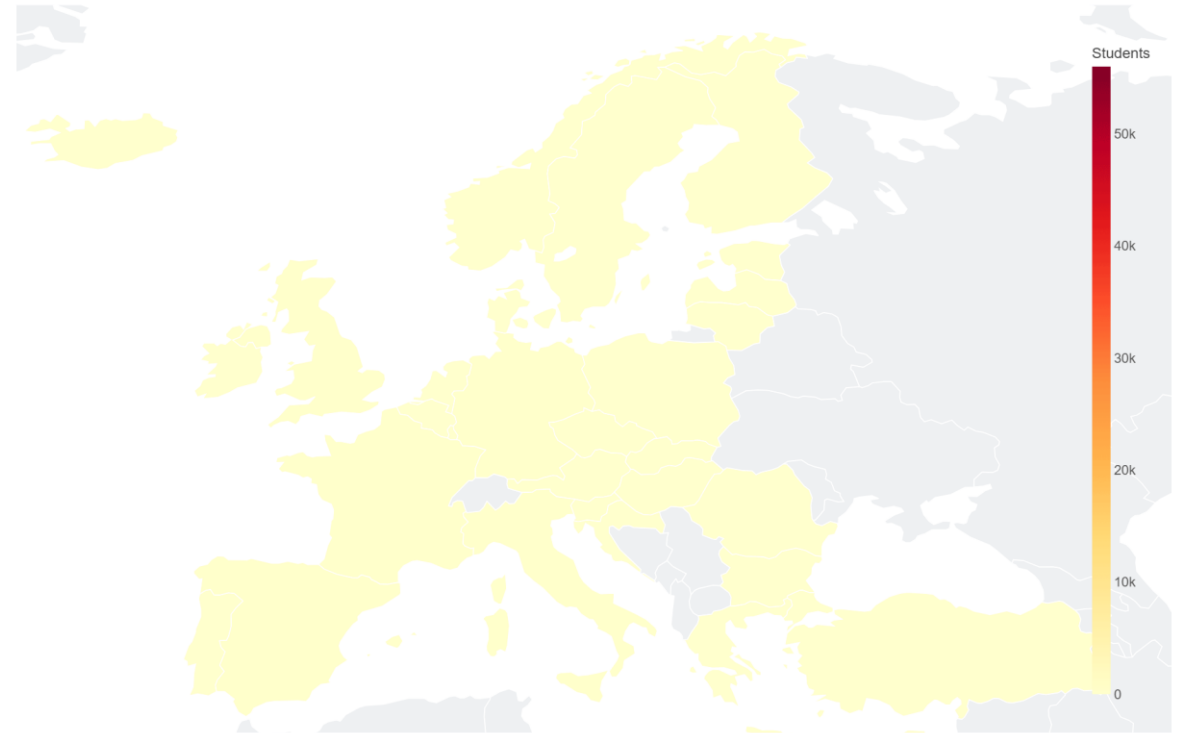


Choropleth Maps 2013-2023

Outgoing Erasmus Students per Country A.Y. 2013



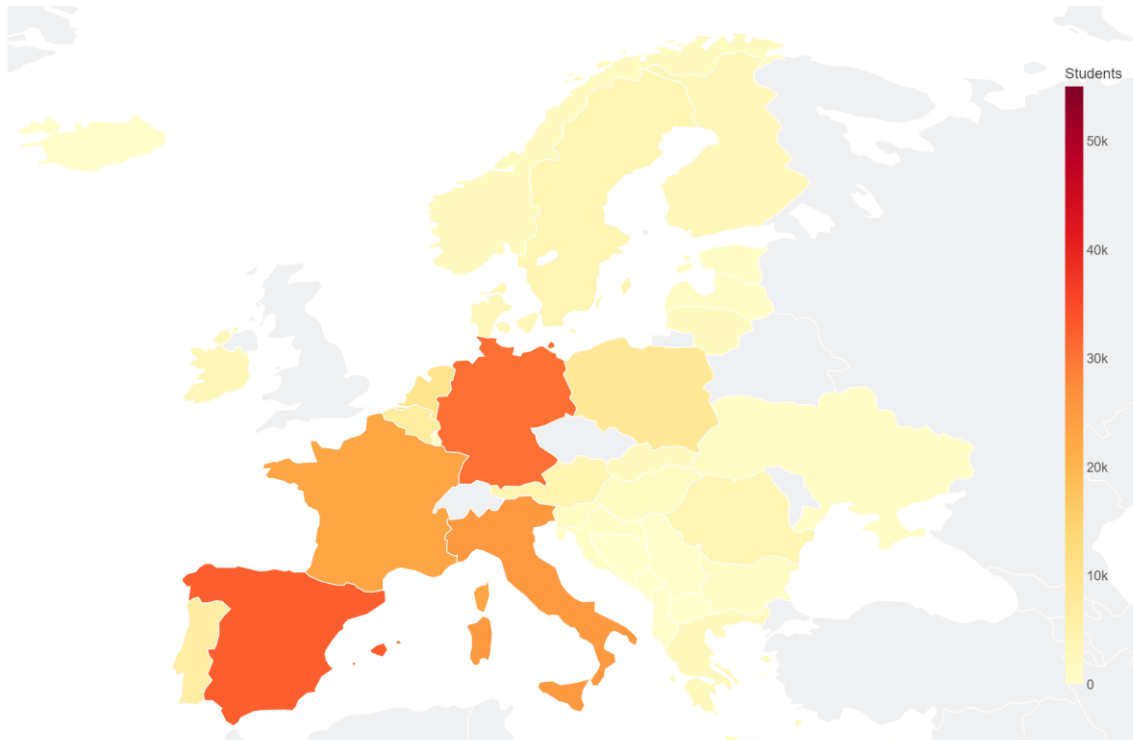
Incoming Erasmus Students per Country A.Y. 2013



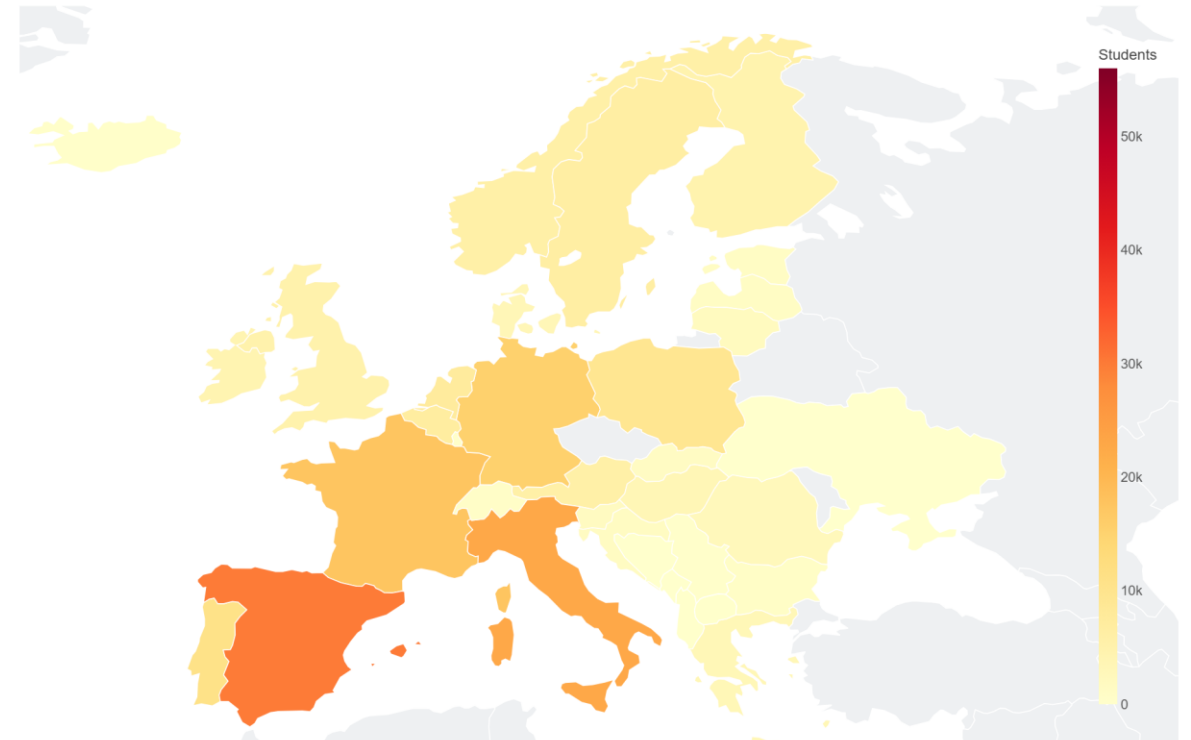
Choropleth Maps 2024

The data for 2024 is incomplete, as it contains only the first semester, but it still allows us to grasp some patterns, such as students from Italy and France favouring going on an exchange in the second semester of the year rather than the first.

Outgoing Erasmus Students per Country A.Y. 2024

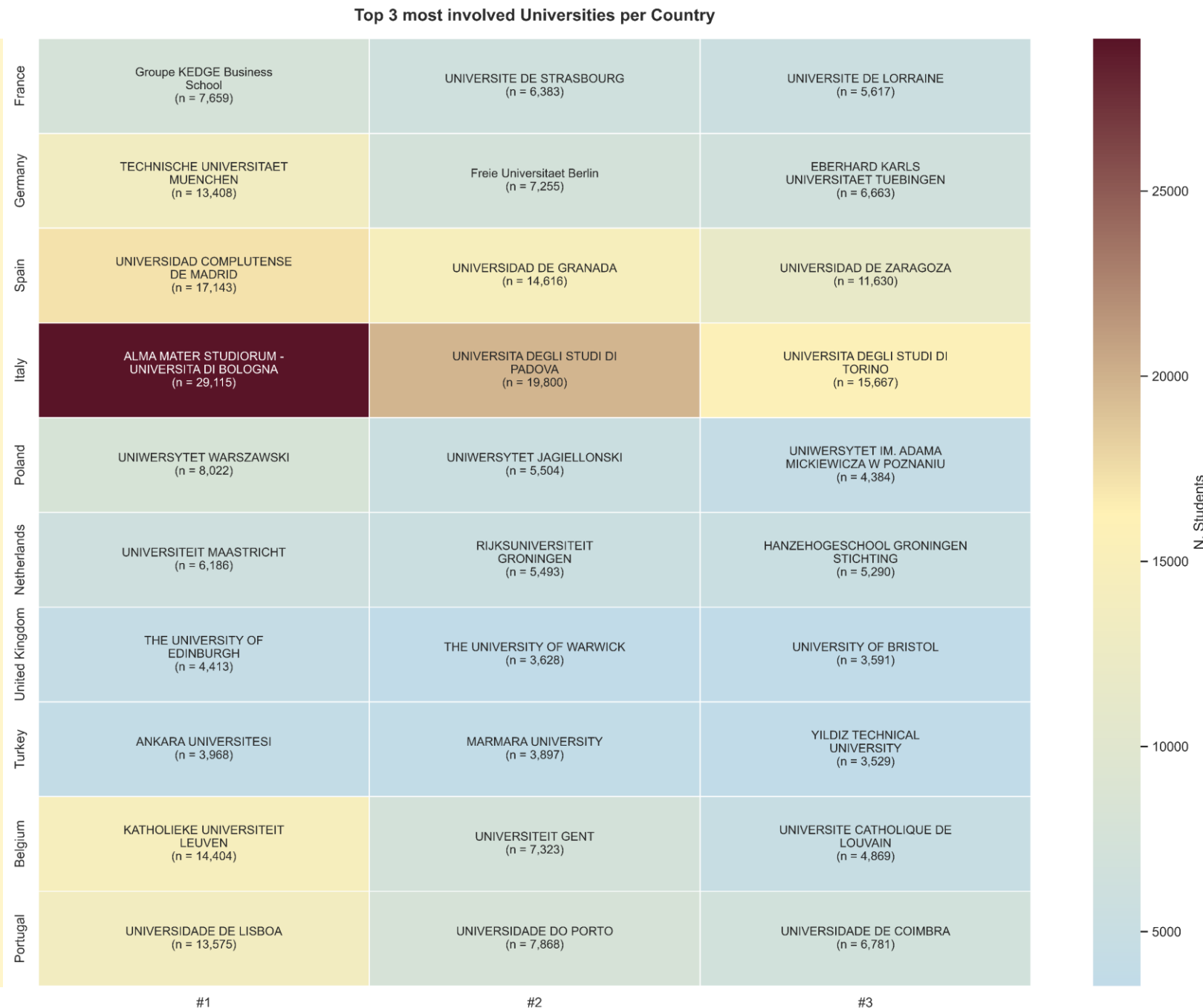


Incoming Erasmus Students per Country A.Y. 2024



Participation Heatmap

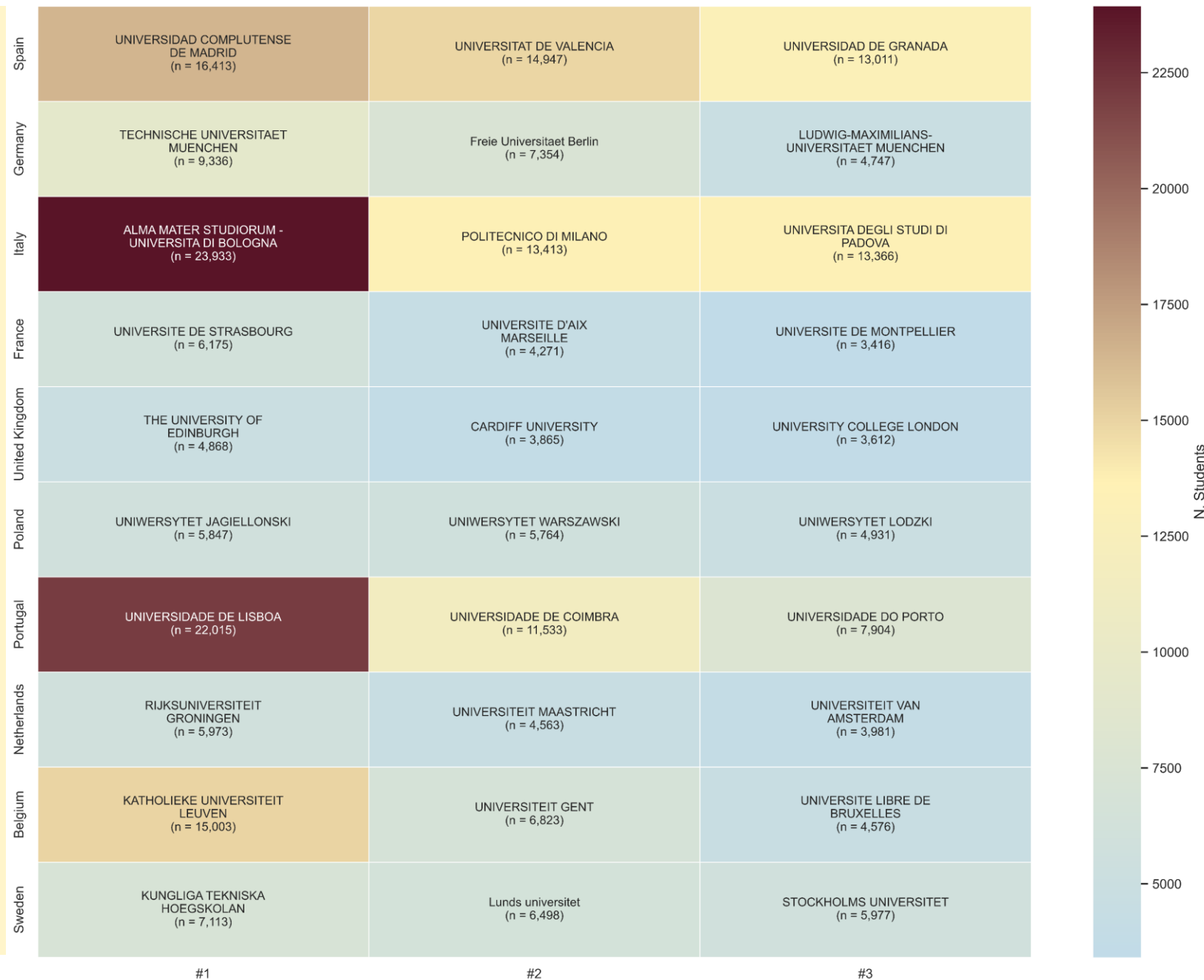
France, despite being the first in terms of participants, shows a lower **average number of participants per institution** compared to other countries.



Attractiveness Heatmap

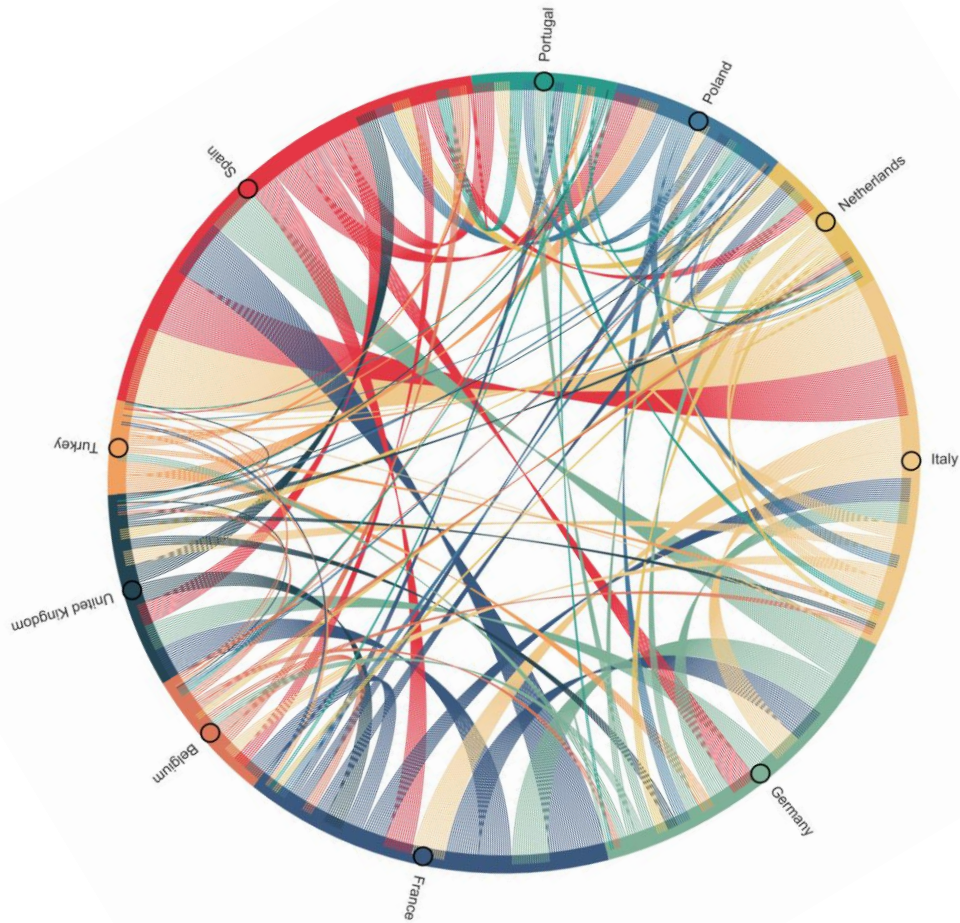
Countries from Southern Europe have significantly higher incoming students compared to the rest of the partners.

Top 3 most attractive Universities per Country

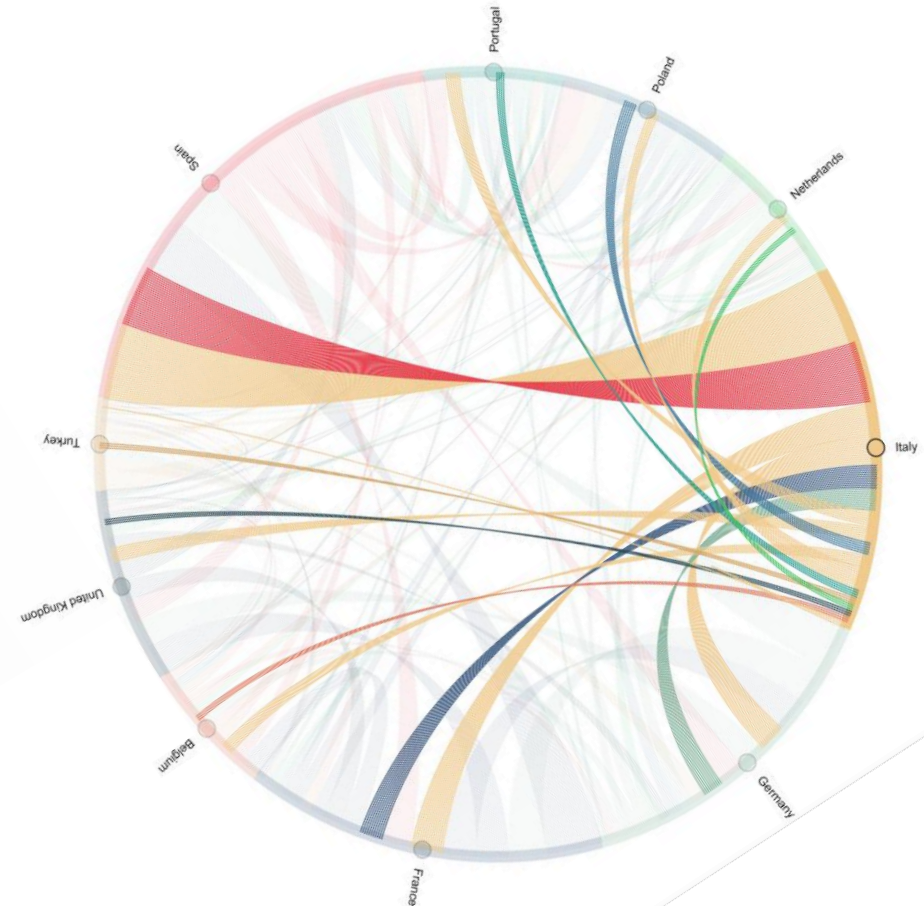


Chord Graph

How the top 10 countries interact within
the Erasmus network



All Europe



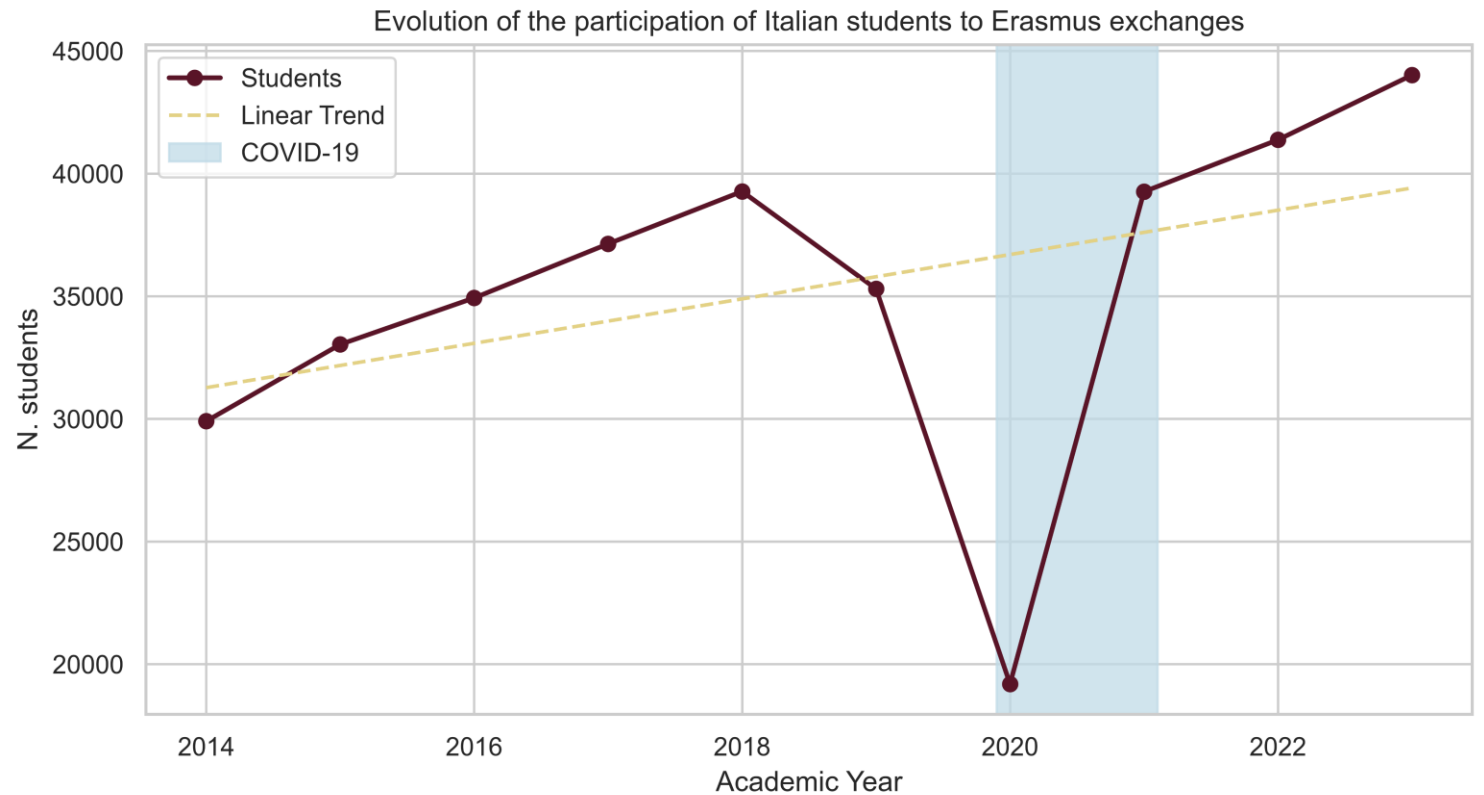
Only Italy

ITALY



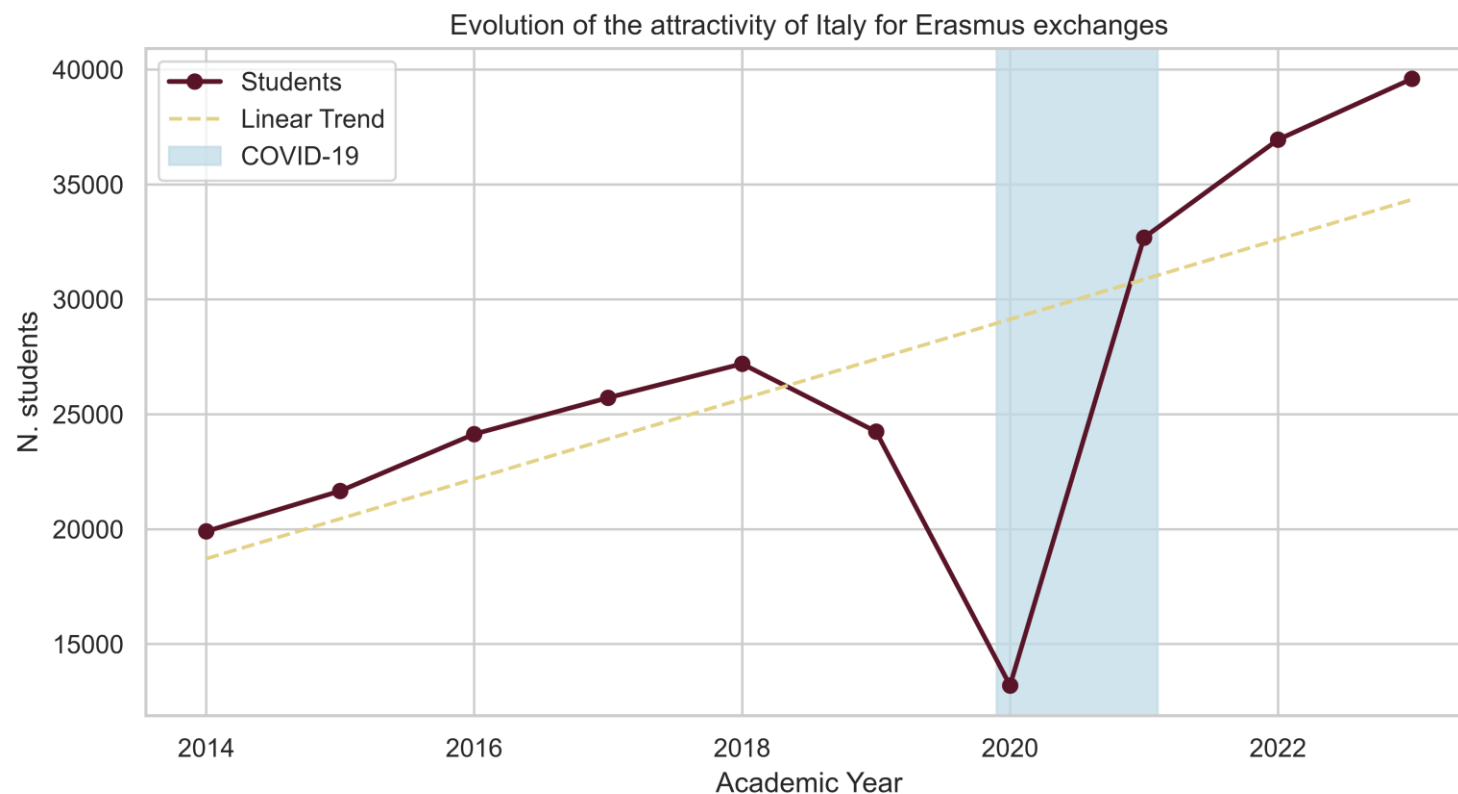
Italian participation

Over the past decade Italy has seen significant growth, increasing the number of outbound exchanges by almost half.



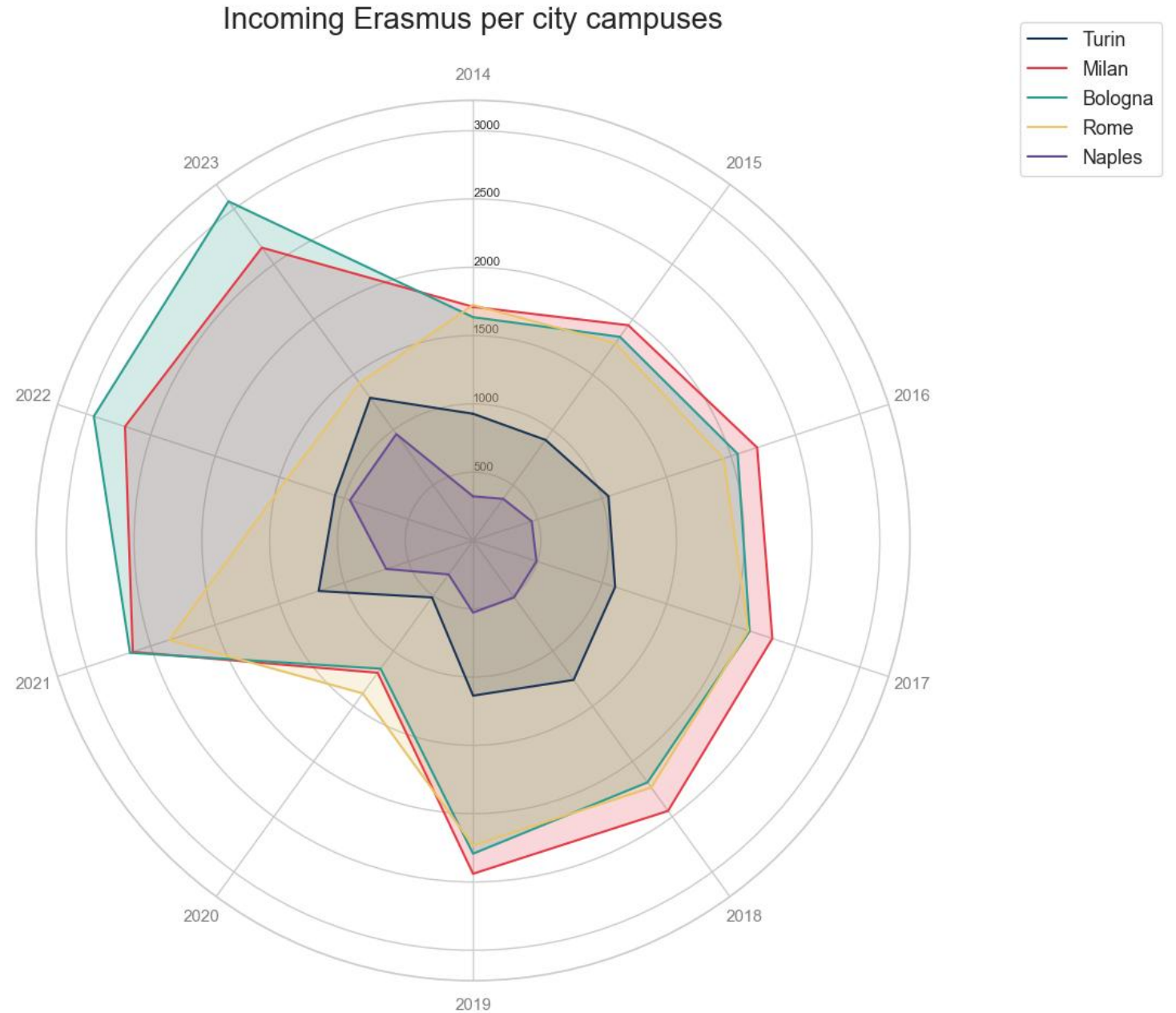
Italy's attractivity

Italy has shown growing numbers with respect to its attractiveness as well, being the third most chosen country in the programme.



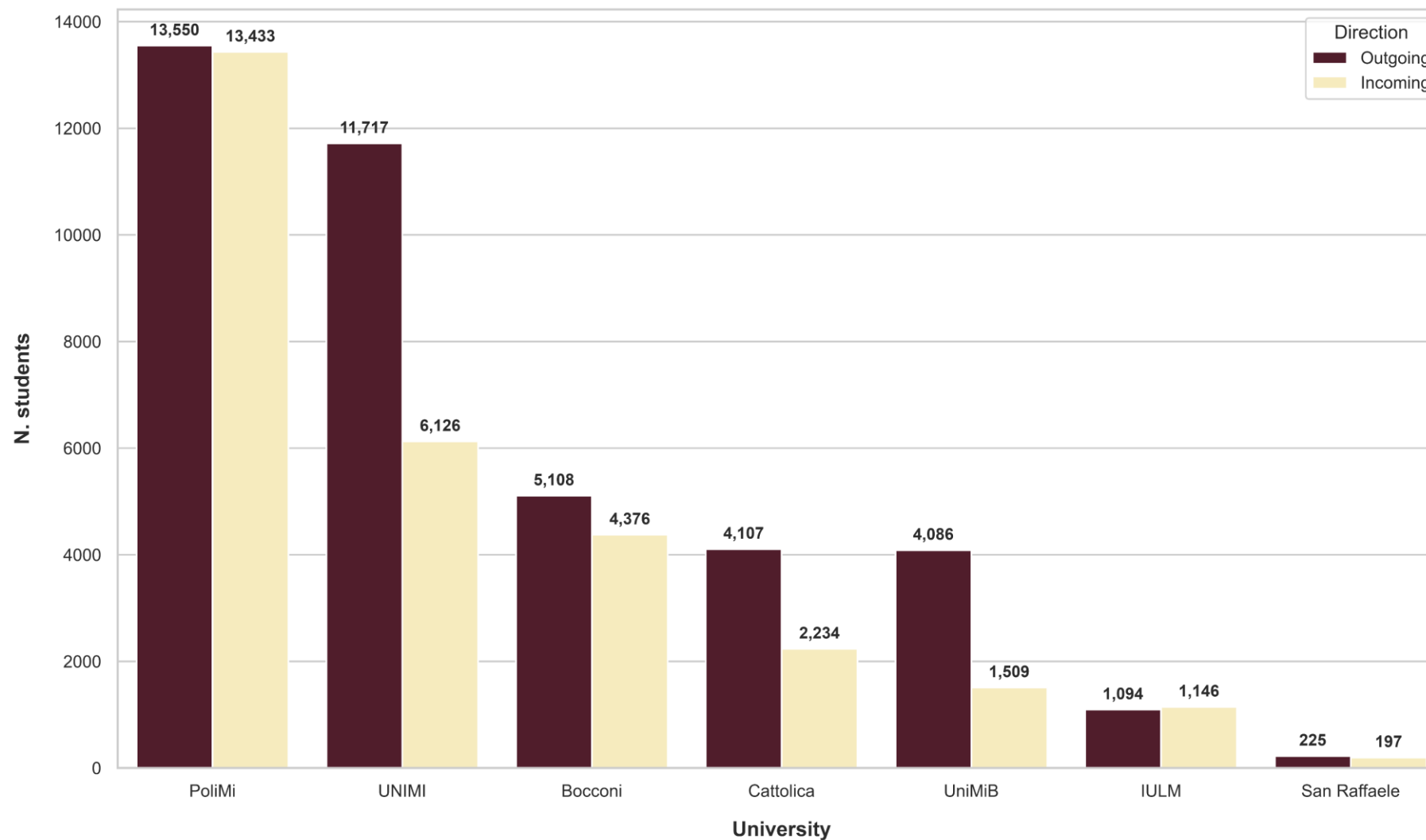
A spider plot is used to better understand how incoming mobility is distributed across the country.

A spider plot is used to better understand how incoming mobility is distributed across the country.



Analysis of Universities of Milan

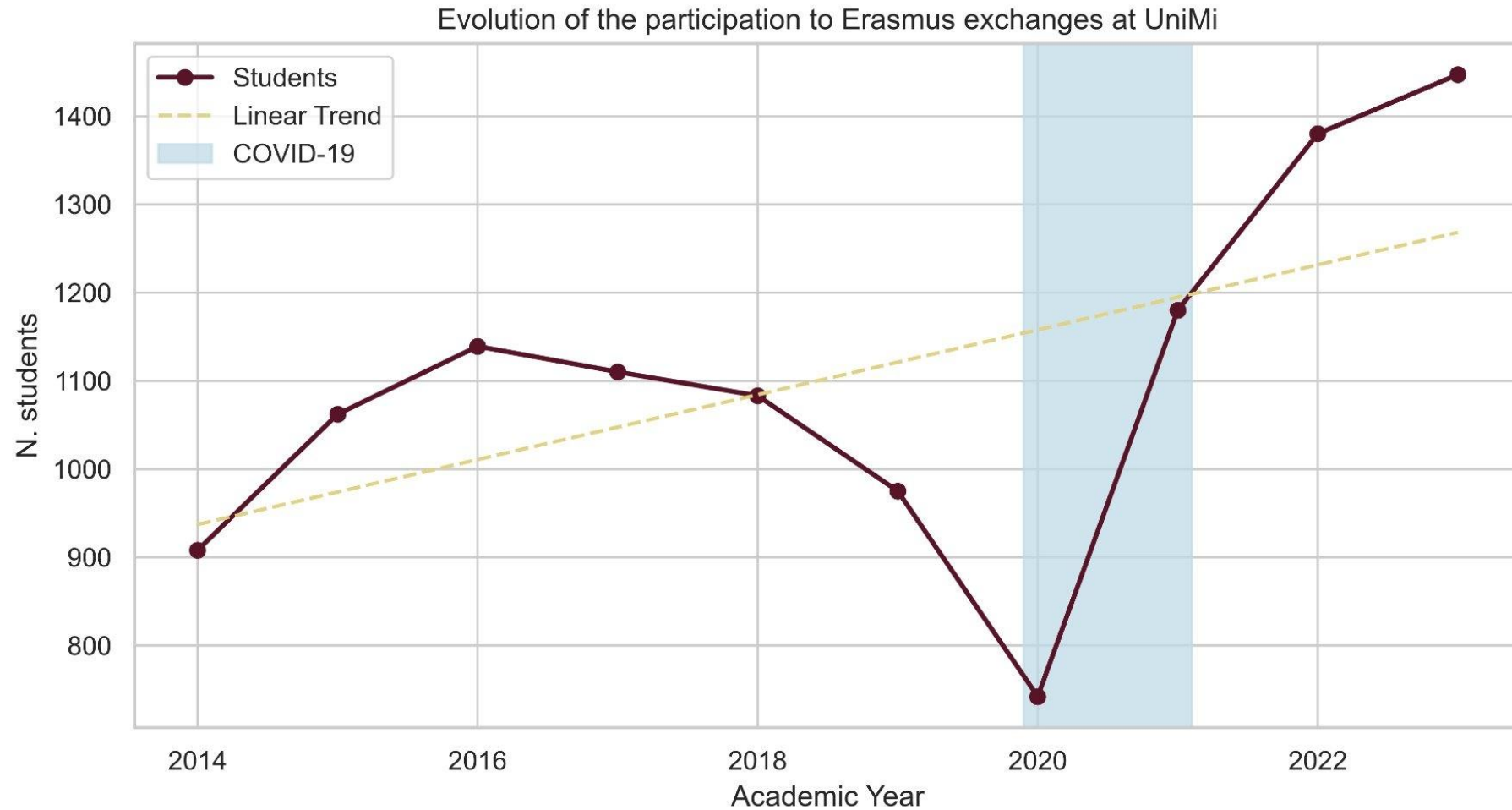
Top 7 Universities of Milan with respect to Erasmus Exchanges



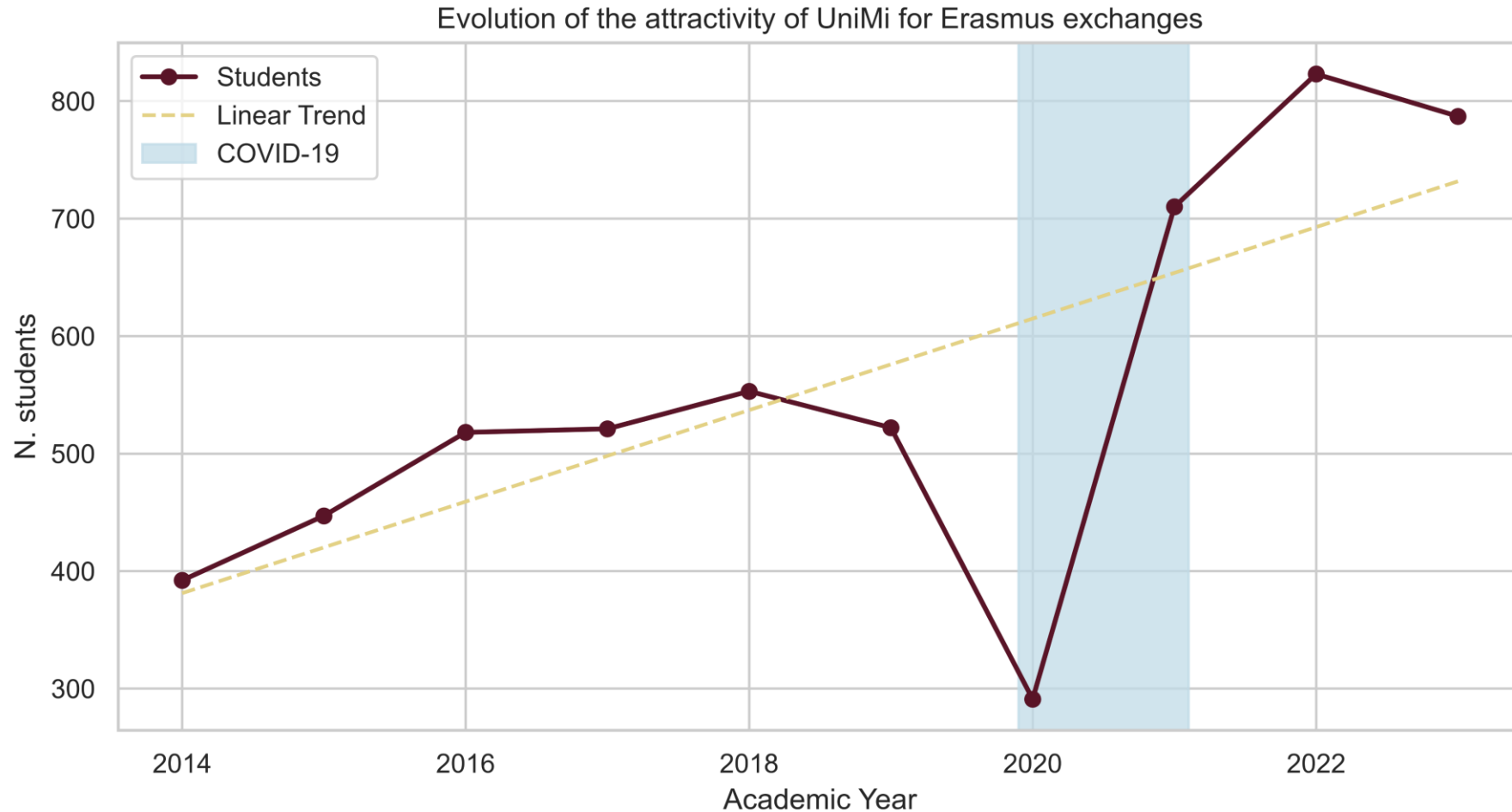
UNIVERSITY OF MILAN



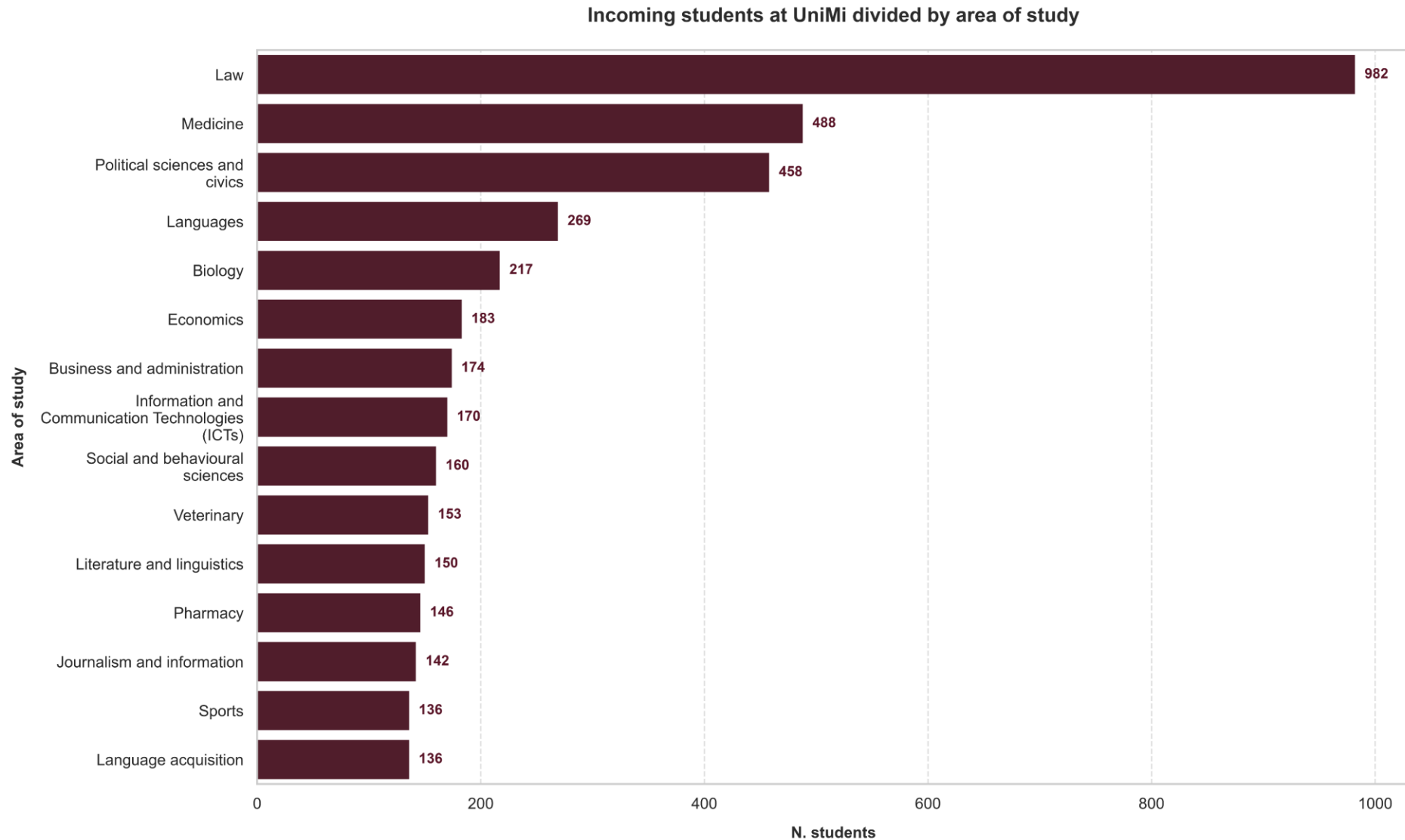
Health of Erasmus – UniMi outgoing



Health of Erasmus – UniMi incoming



Most attractive area of study - UniMi



DATA EXPLORATION



What else can we see from the data?

- 1. Correlations between features**
- 2. Identifying types of exchanges with clustering**

Which insights lie hidden in the data?

1. Correlations between data

We focused on **Mobility Duration** and compared it against two features:

1. **Education Level (ISCED)**

- Spearman correlation
- One-way ANOVA
- Welch's *t*-test

2. **Field of Education**

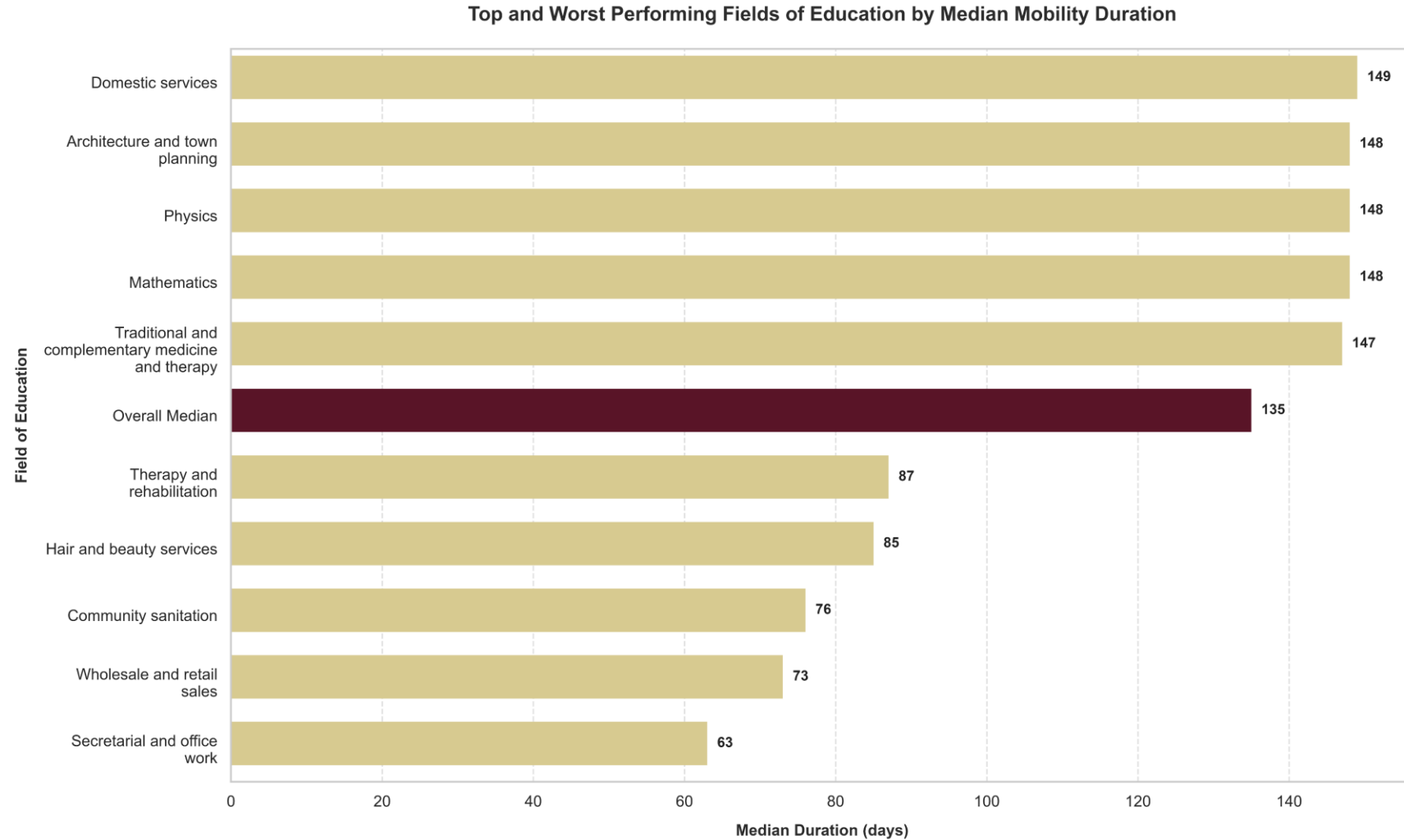
- One-way ANOVA
- η^2

Mobility Duration – Education Level

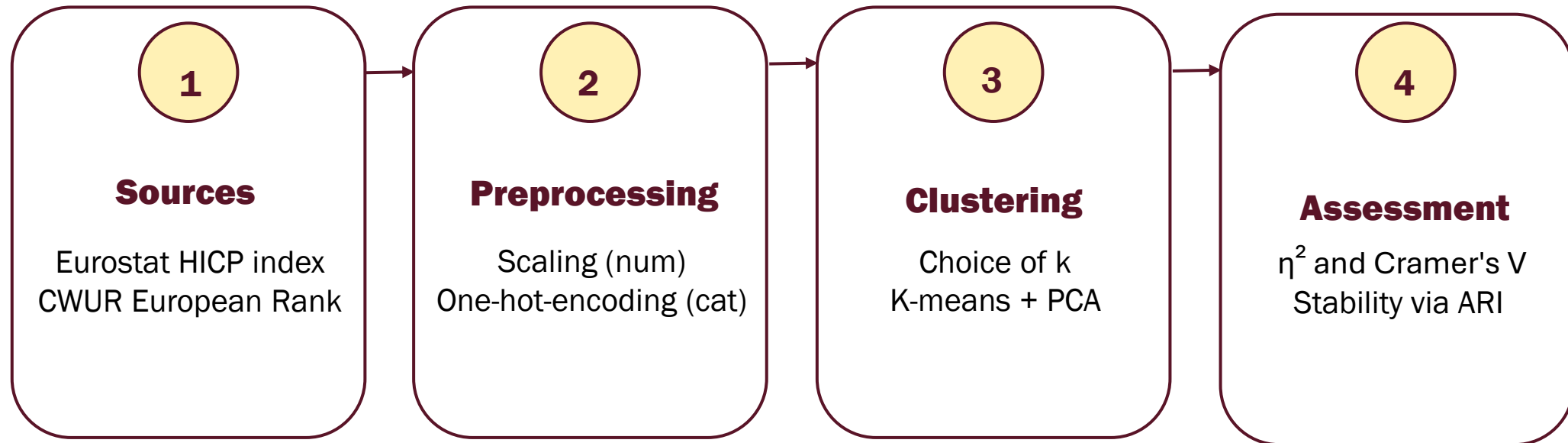
*The statistical tests indicate **no monotonic correlation**, but clear differences across groups and between the two means*

Statistical test	Value	P-value
<i>Spearman</i>	$\rho = 0.0057$	$p \approx 0$
<i>Anova</i>	$F = 24.069$	$p \approx 0$
<i>Welch t-test</i>	$T = 171.27$	$p \approx 0$

Mobility Duration – Field of Education



Confusion Matrix

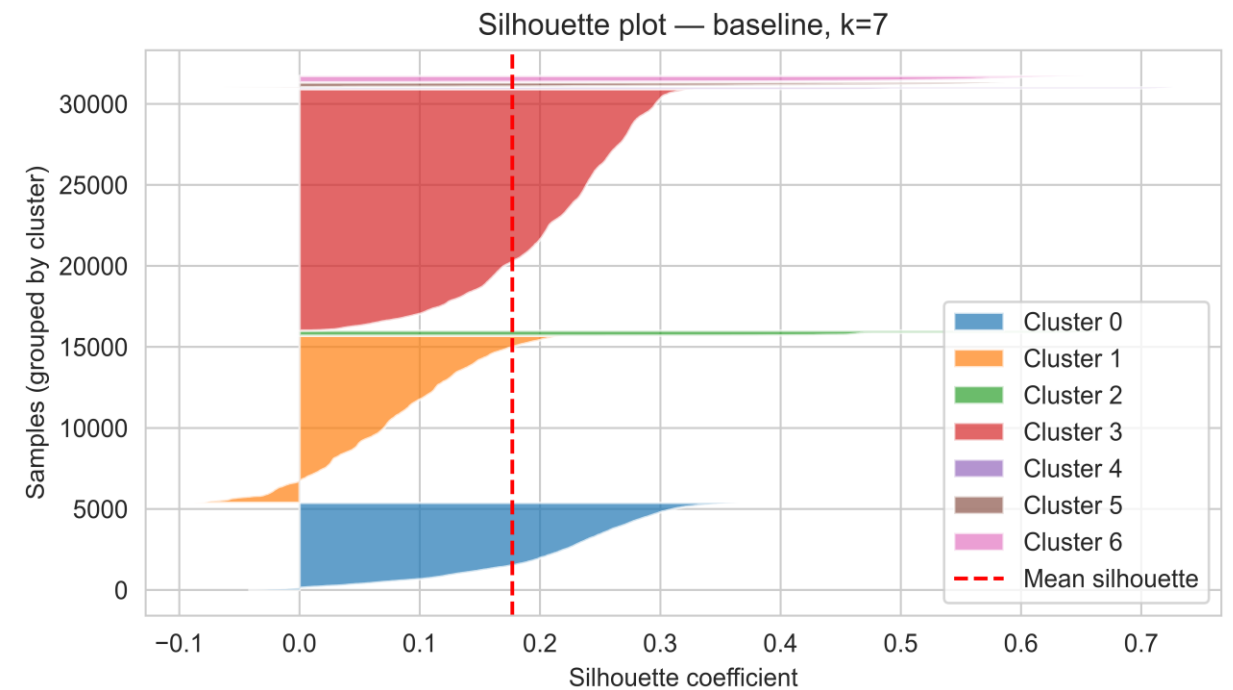
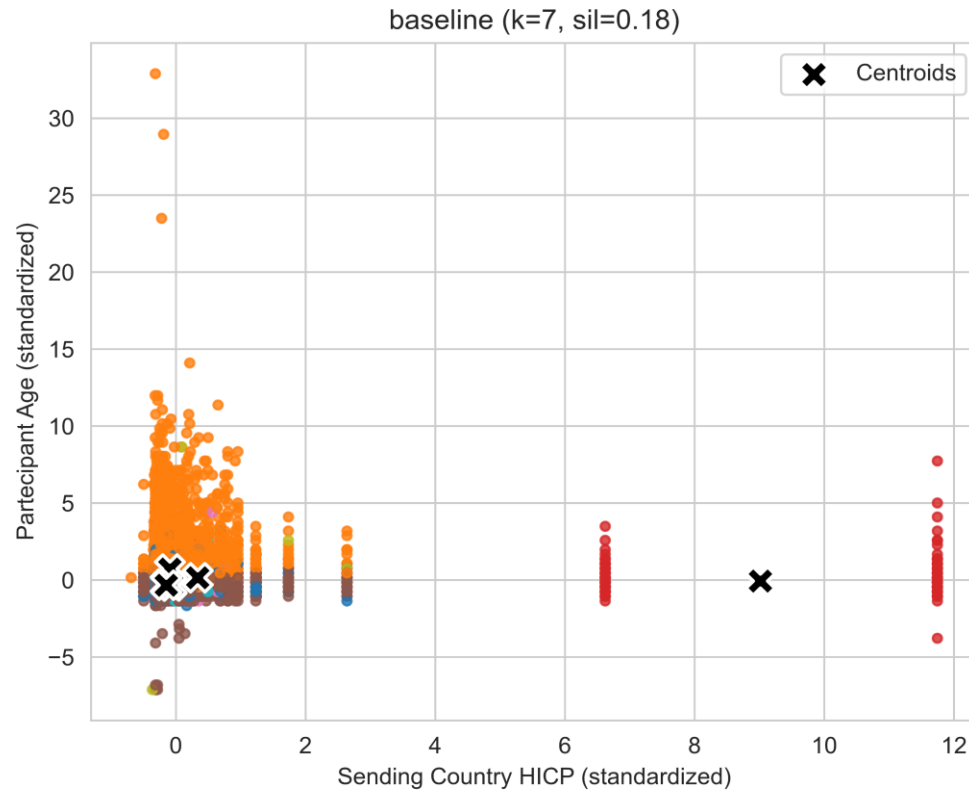


We clustered the dataset by student profiles, highlighting personal information (e.g., **Field of Education**), university-related attributes (**European Ranking**), and economic indicators (**HICP**).

K-means + PCA results

Best $k = 7$

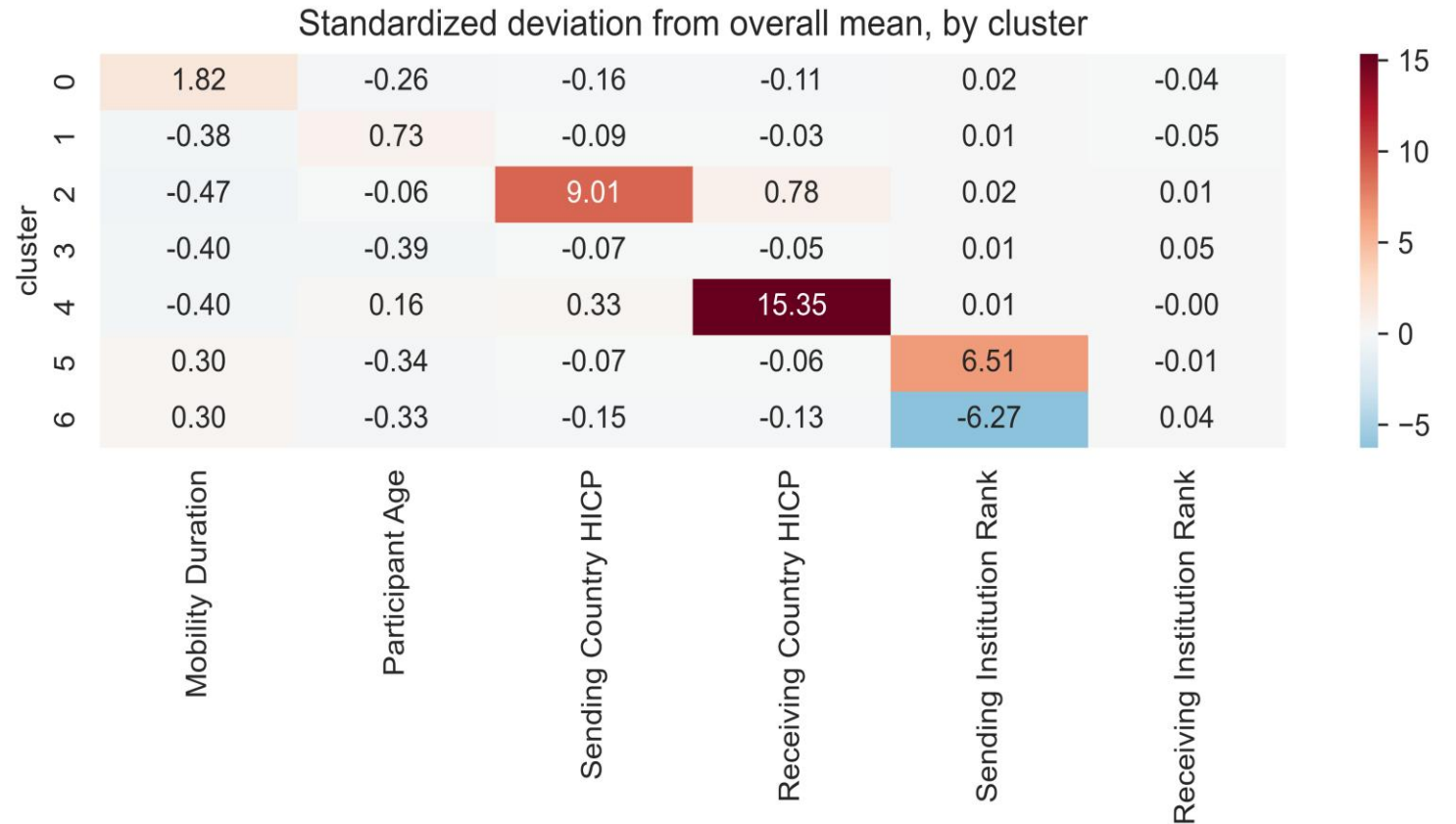
Silhouette score = 0.177 ... pretty rough



a rather confusing view ...

Numerical Feature

Standardized numerical feature deviations highlight the variables that most clearly distinguish each cluster



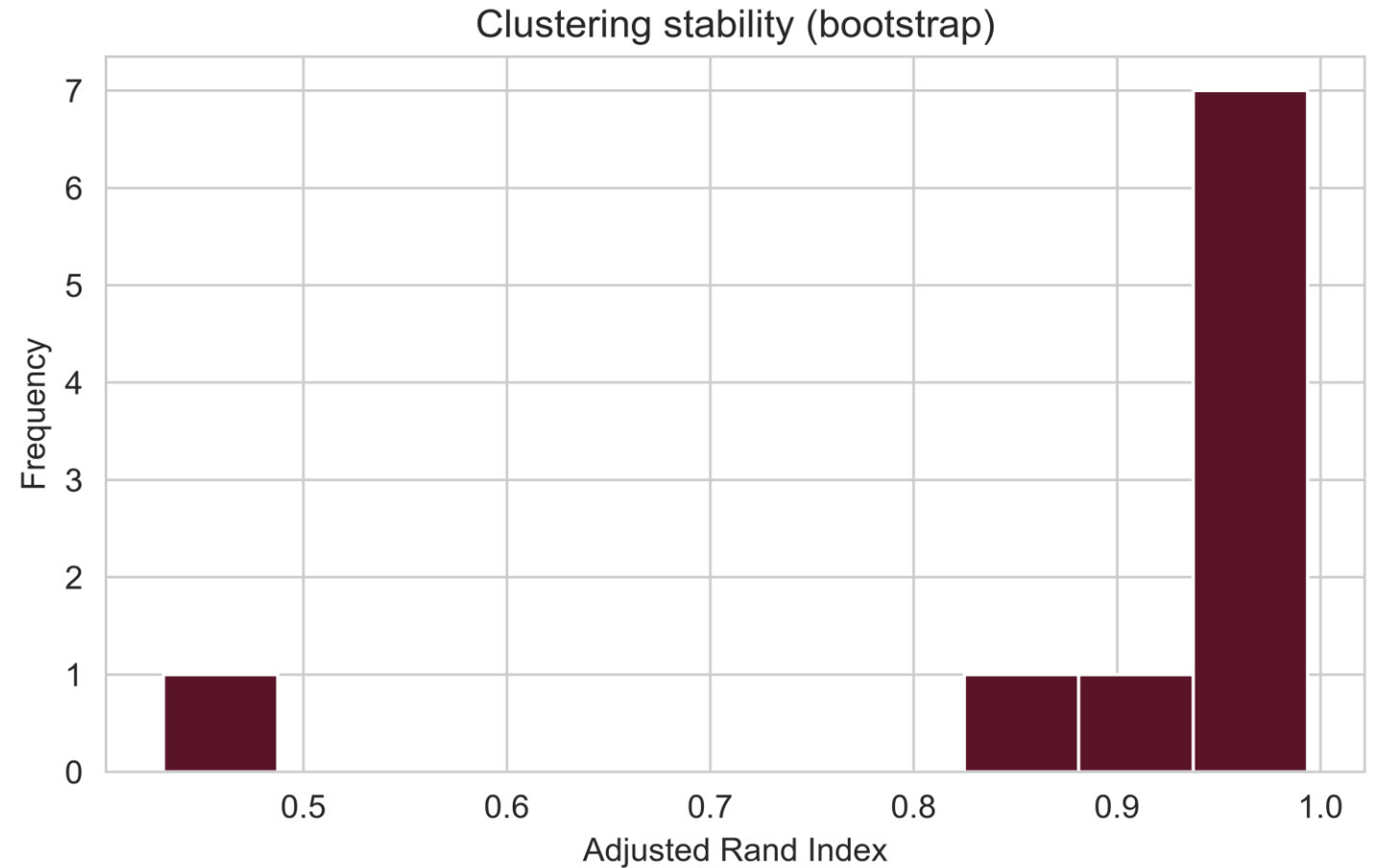
Feature Relevance

Numerical Feature	Type	Effect Size
Sending Institution Rank	<i>numerical</i>	<i>large</i>
<i>HICP diff</i>	<i>numerical</i>	<i>large</i>
<i>Sending Country HICP</i>	<i>numerical</i>	<i>large</i>
<i>Receiving Country HICP</i>	<i>numerical</i>	<i>large</i>
<i>Mobility Duration</i>	<i>numerical</i>	<i>large</i>
<i>Education Level</i>	<i>categorical</i>	<i>medium</i>
<i>Participant Age</i>	<i>numerical</i>	<i>large</i>
<i>Field of Education</i>	<i>categorical</i>	<i>small</i>
<i>Academic Year</i>	<i>categorical</i>	<i>negligible</i>
<i>Fewer Opportunities</i>	<i>categorical</i>	<i>negligible</i>
<i>Participant Gender</i>	<i>categorical</i>	<i>negligible</i>
<i>Receiving Institution Rank</i>	<i>numerical</i>	<i>negligible</i>

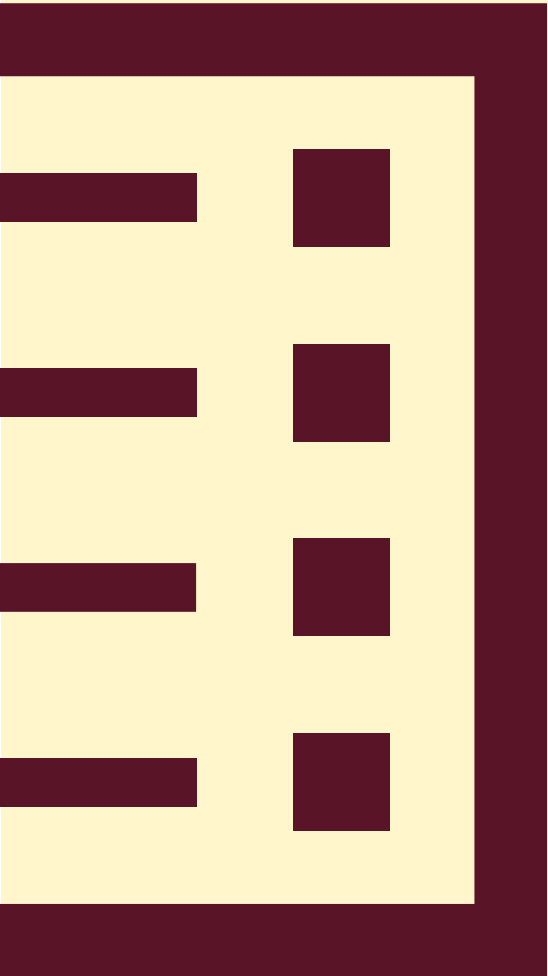
Stability

The bootstrap analysis (ARI) indicates that the clustering is generally stable

Cluster assignments remain consistent across most resampled datasets



DESTINATION PREDICTION

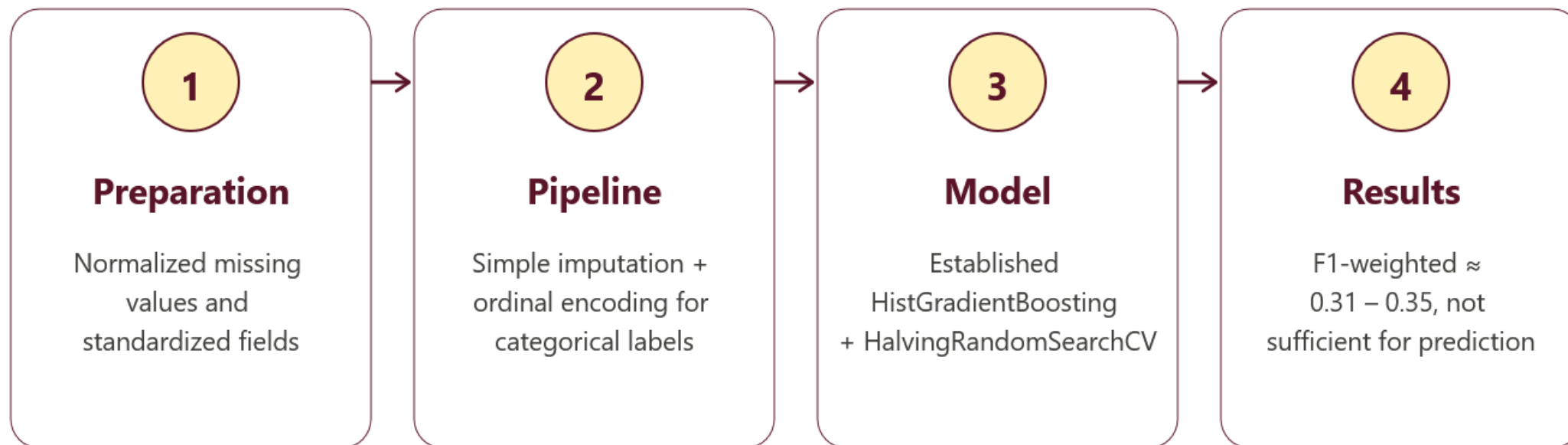


Can we predict the country a student will exchange to?

- Classifier based on the exchanges of previous years
- If not possible, is there anything else we can do?

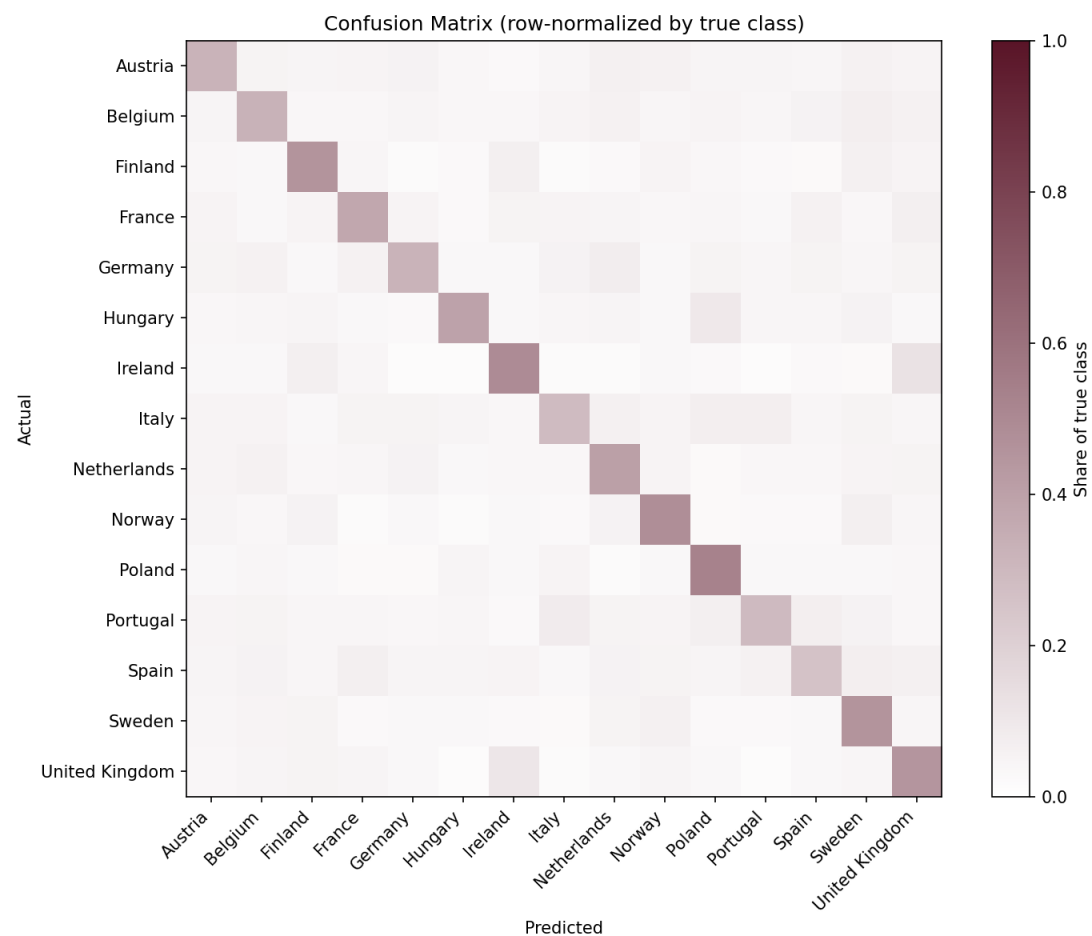
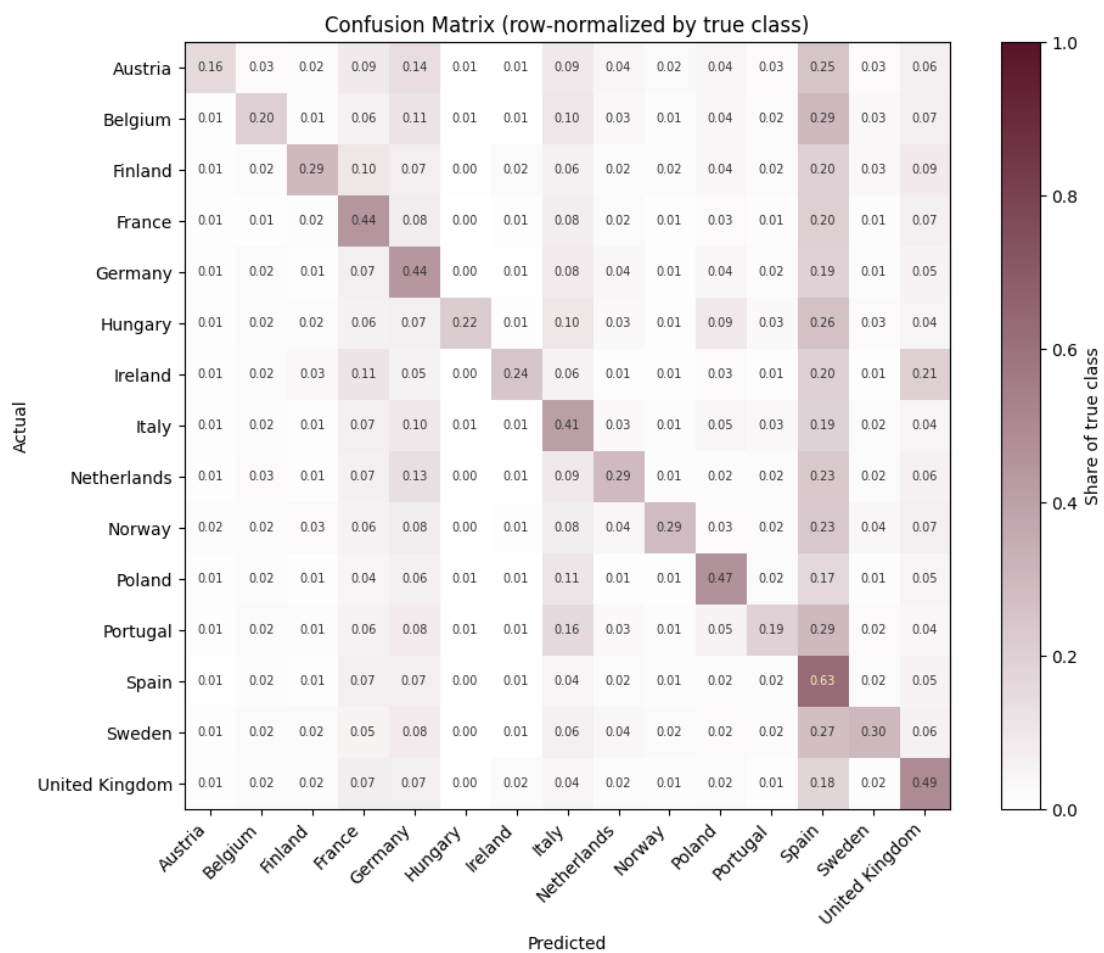
An easy way to pick a destination!

Prediction pipeline



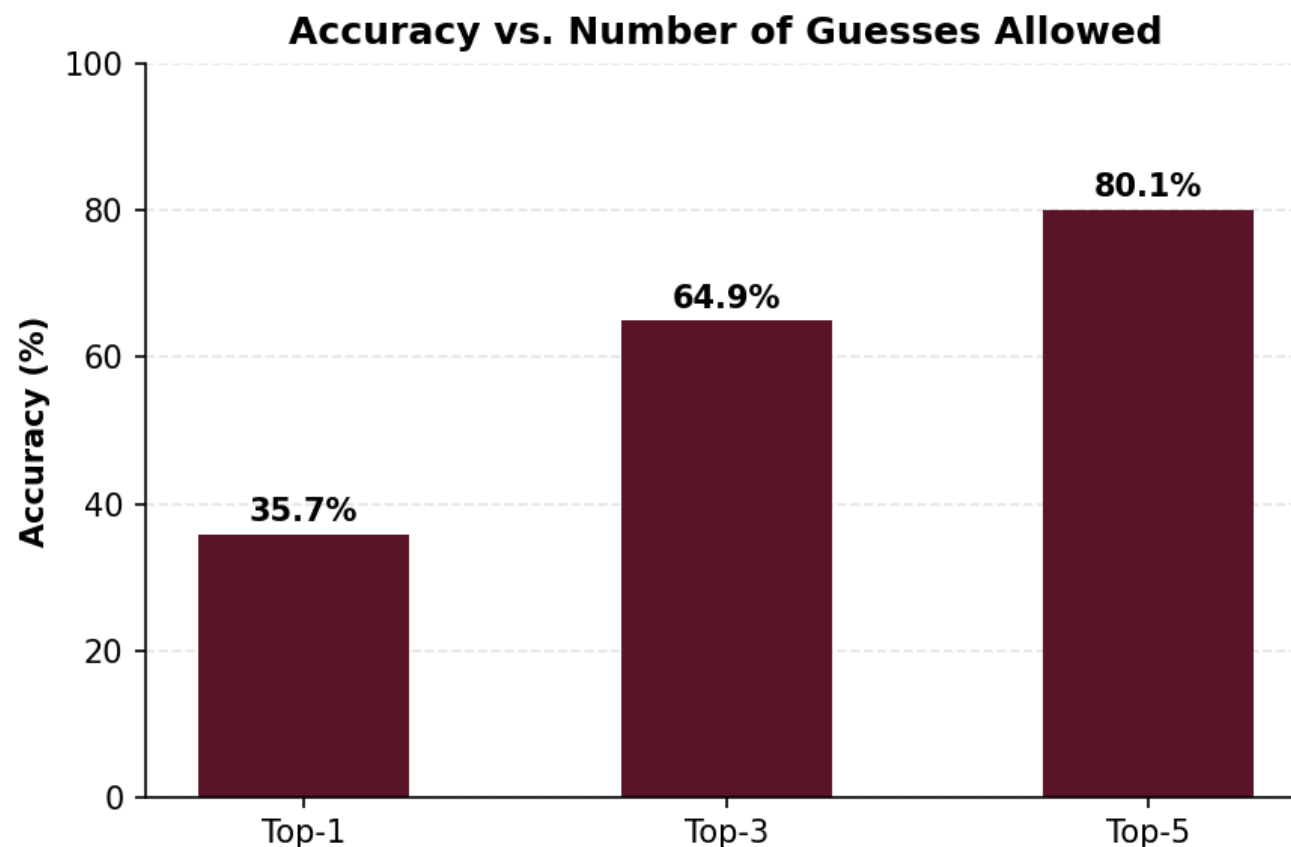
Destination choice is likely driven by factors outside the dataset, e.g. spot availability

Confusion Matrix



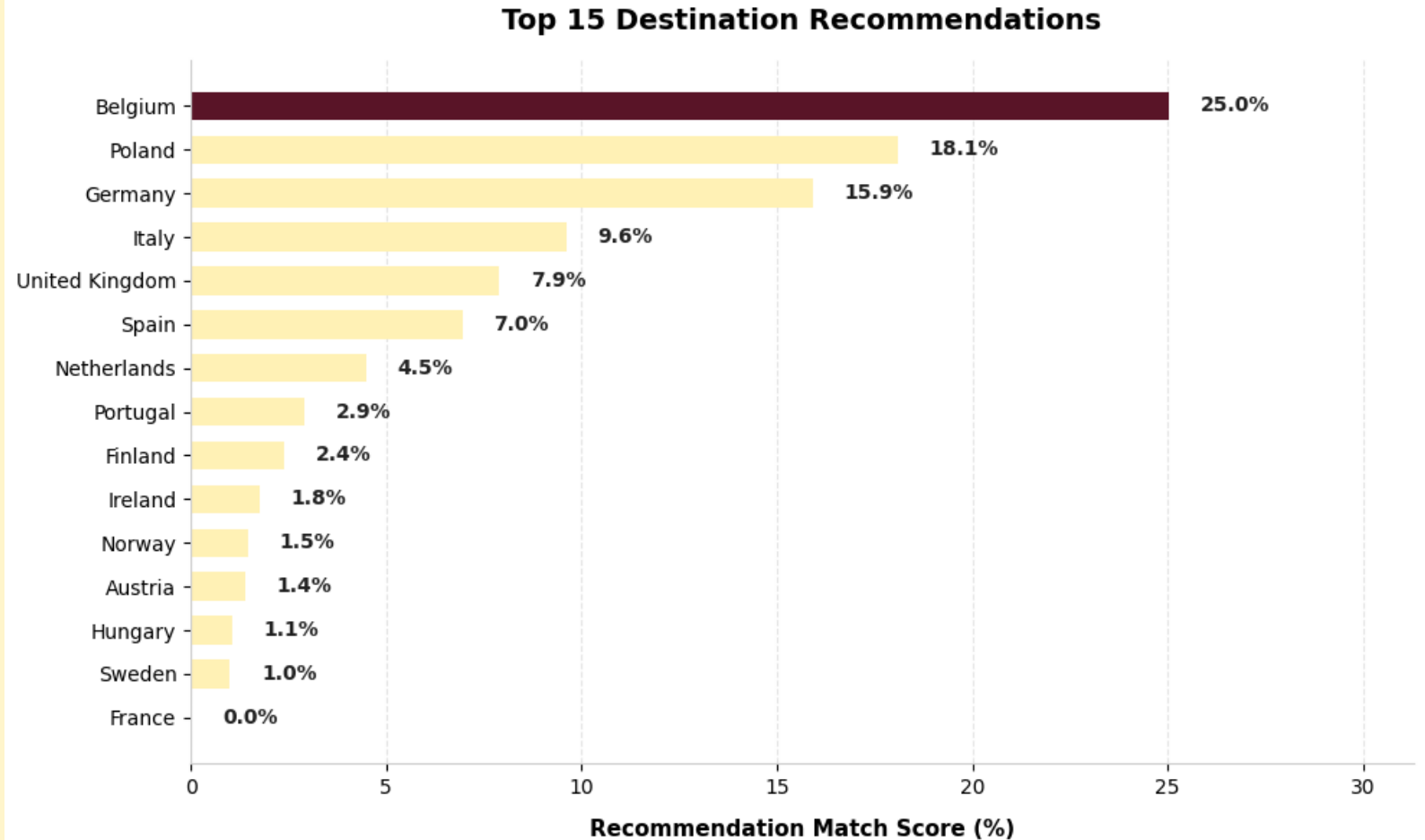
Top K Accuracy

Although the first prediction is mostly wrong, what if we were to increase the number of chances to get it right?



Recommender

Rather than an exact prediction, we get a shortlist of suggested destinations!



CONCLUSION and FUTURE IMPROVEMENTS

The Erasmus+ programme has seen relevant growth in the first ten years of its existence, and despite having to face its fair share of hardships doing Covid, is doing better than ever.

Future improvements

- Compative analysis between areas of studies
- Improving the mappings
- Waiting a few years

Well done Erasmus+!

**Thank you for
your attention!**

